

8 June 2026

To the Editor-in-Chief

Journal of Applied Probability / *Advances in Applied Probability* (Applied Probability Trust)

Dear Editor,

I am pleased to submit the manuscript “**Optimal strategies in the all-heads coin game**” for consideration as a research article in the Applied Probability journals. I leave the assignment between the *Journal of Applied Probability* and *Advances in Applied Probability* to the Editors’ discretion.

Summary of the contribution. The paper analyses a sequential coin-flipping game with two parameters: p , the probability of showing heads, and n , the number of coins. We give the winning probability and the optimal strategy within that game for $p \geq 1/2$ or $p = \frac{1}{2} - \delta$ for small $\delta > 0$.

Fit with the journal. The work is in applied probability: a finite-horizon Markov decision process, its Bellman recursion, the optimality of an explicit stationary policy, and the asymptotics of the value sequence.

Disclosure on the use of generative AI. In line with the Applied Probability Trust’s and Cambridge University Press’s policy on generative AI, the manuscript discloses openly (in the note on authorship in Section 1 and the development history in Appendix A) that the drafting of prose, the Lean proof tactics, and the numerical scripts were produced with Anthropic’s Claude under my direction. The tool is not, and cannot be, listed as an author. I have reviewed every line and take full responsibility for the manuscript; this responsibility is additionally backed by the machine-checked Lean formalization.

An unusual feature: A full formal verification. Note that every numbered theorem, proposition, lemma, and corollary of the paper has been independently machine-checked in Lean 4 / Mathlib. The formalization is structured so that it can be assessed by a referee who is *not* a Lean expert (see Appendix A); a referee may also re-run the verification on the public repository.

Code and data availability. The Lean 4 / Mathlib formalization and the numerical code supporting the paper are openly available in the public repository *coins* (<https://github.com/pfaffelh/coins>) and are permanently archived on Zenodo with DOI 10.5281/zenodo.20489409, as stated in the manuscript’s data availability statement.

Thank you for considering this submission. I look forward to hearing from you.

Sincerely,

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